Cumulative growth of corner cuts in fixed dimension

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Working mathematical draft. The proof has undergone an internal audit; the priority assessment and submission review remain incomplete.

Abstract

Let $C_d(n)$ count the n-point subsets of the nonnegative integer lattice that can be strictly separated from their complements by an affine hyperplane. For every fixed $d \geq 2$, we prove

$$\sum_{j=1}^{n} C_d(j) = \Theta_d\big(n^d (\log n)^{d-1}\big).$$

The upper bound follows from Wagner's estimate. His lifting lemma then gives $C_d(n) = \Omega_d(n^{d-1}(\log n)^{d-2})$ for $d \geq 4$, improving the lower logarithmic exponent in that published baseline. The construction counts primitive lattice simplices in separated boxes, converts their affine spans to vertices of a weight-space arrangement, and sums over disjoint aspect ranges. A finite divisibility argument provides uniform constants without an interchange of density limits and infinite unions. The exact-size three-dimensional gap is not resolved.

1 The counting problem and its scope

A nonempty finite set $D \subset \mathbb{Z}^d_{\geq 0}$ is a *corner cut* if

$$D = \{x \in \mathbb{Z}^d_{\geq 0} : w \cdot x < t\} \quad \text{for some } w \in \mathbb{R}^d_{>0},\ t > 0,$$

with no lattice point on the cutting hyperplane. Write $C_d(n)$ for the number of such sets of cardinality n, and put

$$A_d(n) = \sum_{j=1}^{n} C_d(j).$$

We count sets, rather than the orders of their elements or the weights representing them. Every cut is an order ideal for the coordinate order. In particular, every coordinate of an element of an n-point cut is at most $n-1$, so these counts are finite.

These objects occur as the exponent sets of initial segments of smooth numbers. For distinct primes $p_1, \ldots, p_d$, comparing the products $\prod_i p_i^{x_i}$ is equivalent to comparing $\sum_i x_i \log p_i$. An exponent prefix is consequently a corner cut. Our theorem concerns the full geometric class; it does not give a prime-distribution estimate or a faster random-access algorithm for a fixed prime set.

Corner cuts were introduced in connection with state polyhedra and Gröbner bases by Onn and Sturmfels [1]. The planar order $C_2(n) = \Theta(n \log n)$ is due to Corteel, Rémond, Schaeffer and Thomas [2]. For fixed d, Wagner [3] proved

$$\Omega_d(n^{d-1} \log n) \leq C_d(n) \leq O_d(n^{d-1}(\log n)^{d-1}). \tag{1}$$

His thesis [4, Chapter 3] gives the lower and upper arguments and suggests seeking greater completion multiplicity. Our new step is a lower bound for bounded cardinality, obtained by counting independent aspect ranges. Lifting is used only after this step.

Theorem 1. *For each fixed integer $d \geq 2$,*

$$A_d(n) = \Theta_d\big(n^d (\log n)^{d-1}\big).$$

More explicitly, let

$$R_0(d) = 4d^2(d+1), \qquad n_0(d) = 2^d 4^{R_0(d)}$$

and

$$\kappa_d = \frac{1}{8^{d-1}(100d)^{d^2}(d-1)![4(d+1)]^{d-1}2^{d(d+2)}}.$$

Then, for every integer $n \geq n_0(d)$,

$$A_d(n) \geq \kappa_d n^d (\log_4 n)^{d-1}. \tag{2}$$

Corollary 2. *For each fixed $d \geq 4$,*

$$C_d(n) = \Omega_d\big(n^{d-1} (\log n)^{d-2}\big).$$

This leaves one logarithmic factor between the lower bound and (1). In dimension three, the corollary recovers only the earlier lower order $n^2 \log n$. The cumulative theorem also precludes a uniform reduction of the logarithmic exponent in the upper bound; its positive-proportion consequence is stated in Section 5.

2 Primitive simplices in axis boxes

Fix an ambient integer $d \geq 2$ and set $\delta = 1/(100d)$. For integer side lengths $L_1, \dots, L_d$, put

$$m = \min_i L_i, \quad M = \max_i L_i, \quad Q = \prod_{i=1}^{d} L_i.$$

Define $B_i \subset \mathbb{Z}^d_{\geq 0}$ by taking the jth coordinate interval to be

$$[L_j, L_j + \lfloor \delta L_j \rfloor] \quad (j = i), \qquad [0, \lfloor \delta L_j \rfloor] \quad (j \neq i).$$

A d-by-$(d-1)$ integer matrix is called primitive if the greatest common divisor of its maximal minors is one.

Lemma 3. *Assume $M \leq m^2$ and $m \geq 2^{8d^2}$. At least a fraction $8^{-(d-1)}$ of the ordered choices $p_i \in B_i$ have primitive difference matrix*

$$V = [p_2 - p_1 \ \cdots \ p_d - p_1].$$

The minor criterion and its equivalence with full column rank modulo every prime are standard [6, Propositions 3–4]. The point requiring proof here is uniformity for these moving, possibly anisotropic boxes.

Proof. Keep the same δ throughout a dimension induction. Let E_r be the event that the first r points, projected onto the first r coordinates, have primitive difference matrix. The event E_1 is automatic, with empty minor equal to one. At step $2 \leq r \leq d$, fix a primitive

configuration of the previous $r-1$ projected points. Its difference matrix U has size $(r-1)$-by-$(r-2)$. Its vector a_0 of signed maximal minors is primitive, and U has rank $r-2$ modulo every prime.

Let u be the first $r-1$ coordinates of $p_r - p_1$, and put $D = \det[U\ u]$. This determinant is nonzero for every allowed choice. Indeed, the normalized row matrix of the previous $r-1$ projected points differs from the identity by at most $(r-1)\delta$ in maximum row-sum norm. The normal of their affine span, normalized to height one, has scaled coordinates within $(r-1)\delta/(1-(r-1)\delta)$ of one and is positive. The new projected point has every scaled coordinate at most δ; it is strictly on the origin side of that affine span. Also,

$$|D| \le (r-1)!\, 2^{r-1} M^{r-1} \le 2^{d^2} m^{2(d-1)}. \tag{3}$$

Every coordinate interval contains more than δm integers. For a prime ℓ, the equation $D = 0 \pmod{\ell}$ is a nontrivial linear equation in the new projected point: at least one coordinate of a_0 is nonzero modulo ℓ. Its probability is at most $1/\ell + a/m$, where $a = 100d$.

Suppose $\ell \mid D$. Since U has full column rank modulo ℓ, there is a unique vector λ with $u = U\lambda \pmod{\ell}$. Write z for the rth row of the old columns and u_r for the rth coordinate of the new column. The full r-by-$(r-1)$ matrix loses rank modulo ℓ precisely if

$$u_r = z\lambda \pmod{\ell}. \tag{4}$$

Conditional on everything except the last coordinate of p_r, this is one residue class, of probability at most $1/\ell + a/m$. Thus the probability of failure at any particular prime is at most $(1/\ell + a/m)^2$.

Put $s = \lfloor (\log_2 m)/2 \rfloor$, $P = 2^s$. The small-prime union bound gives

$$\begin{aligned} \Pr(\text{failure at a prime } \ell \le P) &\le \sum_{\substack{\ell \le P \\ \ell \text{ prime}}} (1/\ell + a/m)^2 \\ &\le \frac{3}{4} + \frac{2a(s+1)}{2^{2s}} + \frac{a^2}{2^{3s}}. \end{aligned} \tag{5}$$

Here $\sum_{j=2}^{\infty} j^{-2} < 3/4$ follows by comparison with $1/4 + \int_2^{\infty} x^{-2}\,dx$. The cross term uses $H_P \le 1 + \log P \le 1 + s$, and the constant term uses at most P primes.

For large primes, first fix the new projected coordinates. Every bad prime $\ell > P$ divides the same nonzero integer D. By (3), their number is at most

$$\frac{\log_2 |D|}{s} \le 4(d-1) + \frac{d^2 + 4(d-1)}{s} \le b := 4d - 3 \quad \text{if } s \ge d^2 + 4(d-1).$$

For each of these finitely many primes, (4) bounds the conditional last-coordinate failure probability by $2^{-s} + a2^{-2s}$. Their total contribution is therefore at most $b(2^{-s} + a2^{-2s})$.

Define

$$E_d(s) = \frac{2a(s+1)}{2^{2s}} + \frac{a^2}{2^{3s}} + b(2^{-s} + a2^{-2s}). \tag{6}$$

This decreases for $s \ge 1$. For every integer $d \ge 2$, the choice $S_d = 4d^2$ satisfies $S_d \ge d^2 + 4(d-1)$ and $E_d(S_d) < 2^{-10} < 1/8$. For an explicit verification valid in every dimension, use $a \le 2^{4d}$, $b \le 2^{2d}$ and $4d^2 + 1 \le 2^{3d}$. They give

$$E_d(4d^2) \le 2^{1+7d-8d^2} + 2^{8d-12d^2} + 2^{2d-4d^2} + 2^{6d-8d^2} < 2^{-10}.$$

The four exponents decrease for $d \ge 2$, and at $d = 2$ are $-17, -32, -12, -20$.

Consequently the conditional probability of a nonprimitive extension is at most $3/4 + E_d(s) \le 7/8$. To justify the conditioning across steps, the events $E_1, \dots, E_{r-1}$ depend only

on previous projected coordinates. After these are fixed, the old rth coordinates and the new point retain their product distributions. Thus

$$\Pr(E_r \mid E_1, \dots, E_{r-1}) \geq 1/8.$$

Multiplication proves that $\Pr(E_1 \cap \cdots \cap E_d) \geq 8^{-(d-1)}$. This intersection implies E_d, as required. All bounds on large primes were finite before any limit was taken. □

3 From simplices to cuts

We first recall a useful elementary fact about arrangements, including the case of dependent hyperplanes.

Lemma 4. *For a finite affine-hyperplane arrangement in a nonempty open convex domain in $\mathbb{R}^k$, the number of chambers is at least the number of interior vertices.*

Proof. In dimension one, v distinct interior points give $v + 1$ chambers. For the induction step, discard duplicate hyperplanes and insert the remaining ones in a fixed order. Inserting H creates exactly the number of chambers of the restricted arrangement in H intersected with the domain. This follows because every old chamber is open and convex and is split exactly when it meets H.

Assign each vertex to its last incident hyperplane H. The incident normals span $\mathbb{R}^k$, and the nonzero normal of H extends to a basis among them. Hence the assigned vertex is also an interior vertex of the restriction to H. The induction hypothesis bounds the number assigned to H by the number of new chambers. Summing gives the result. □

Lemma 5. *Under the hypotheses of Lemma 3, at least*

$$\rho_d Q^d, \qquad \rho_d = 8^{-(d-1)}(100d)^{-d^2}, \tag{7}$$

distinct corner cuts are represented by weights in

$$\Omega_L = \prod_{i=1}^{d} \left(\frac{9}{10L_i}, \frac{11}{10L_i} \right),$$

using threshold one. Each has at most $(3/2)^d Q$ points. Its maximum ith coordinate on the ith axis lies in $(4L_i/5, 6L_i/5)$.

Proof. Let $p_i \in B_i$ have primitive difference matrix V. Its primitive cofactor normal is a. There is an integer vector z with $a \cdot z = 1$; the matrix $[V\ z]$ is unimodular, after choosing the cofactor orientation. The columns of V therefore form a basis of $\{x \in \mathbb{Z}^d : a \cdot x = 0\}$.

After row scaling, the submatrix of V in coordinates $2, \dots, d$ differs from the identity by at most $(d-1)\delta$ in maximum row-sum norm. If two integer points in one B_i belong to the selected affine span, their difference is Vc for an integer vector c. The scaled coordinate difference is at most δ, so

$$\|c\|_\infty \leq \frac{\delta}{1 - (d-1)\delta} < 1.$$

Thus $c = 0$. The plane meets each box in exactly its selected point, so distinct primitive choices give distinct planes.

The normalized row matrix of all d points differs from the identity by at most $d\delta$. The unique solution of $p_i \cdot w = 1$ satisfies

$$|L_i w_i - 1| \leq \frac{d\delta}{1 - d\delta} < 1/10.$$

In particular it is positive and lies in Ω_L. Each selected plane gives a distinct interior vertex of the arrangement $x \cdot w = 1$. This arrangement can be taken finite: only nonnegative integer points with $x_i \le \lceil 10L_i/9 \rceil$ for every i can meet Ω_L; all other points have weight greater than one throughout that domain.

Every chamber gives a corner cut. Different chambers give different sets, since a fixed collection of strict signs has convex intersection with Ω_L. By Lemma 4, the number of these cuts is at least the number of selected vertices. Each B_i has at least $\delta^d Q$ points, so Lemma 3 proves (7).

A coordinate of an included point is smaller than $10L_i/9$. There are fewer than $10L_i/9 + 1 \le (3/2)L_i$ possible integer values when $L_i \ge 3$. This proves the cardinality bound. For a chamber weight, the maximum axis coordinate is $\lfloor 1/w_i \rfloor$; the chamber condition excludes equality $1/w_i \in \mathbb{Z}$. It lies strictly between $4L_i/5$ and $6L_i/5$ when $L_i \ge 10$. Both side conditions follow from the stated threshold. □

4 Summing aspect ranges

Proof of Theorem 1. The upper bound follows by summing (1). For the lower bound, take R maximal with

$$Q = 4^R \le n/2^d.$$

Choose any integer tuple with

$$r_1 + \cdots + r_d = R, \qquad r_i \ge \lceil R/(d+1) \rceil, \quad \text{and put } L_i = 4^{r_i}.$$

Then

$$Q^{1/(d+1)} \le L_i \le Q^{2/(d+1)}.$$

In particular $\max L_i \le (\min L_i)^2$. When $n \ge n_0(d)$, we have $R \ge R_0(d)$ and $\min L_i \ge 2^{8d^2}$, so Lemma 5 applies.

All the resulting cuts have at most $(3/2)^d Q < n$ points. No cut is counted for two distinct scale tuples: such tuples differ by a factor of at least four in some L_i, whereas the observable maximum axis coordinate belongs to $(4L_i/5, 6L_i/5)$. These ranges are disjoint.

Write $h = R - d\lceil R/(d+1) \rceil$. There are $\binom{h+d-1}{d-1}$ tuples. For $R \ge R_0(d)$,

$$h \ge R/(d+1) - d \ge R/[2(d+1)],$$

and hence

$$\binom{h+d-1}{d-1} \ge \frac{R^{d-1}}{[2(d+1)]^{d-1}(d-1)!}.$$

Also $Q > n/2^{d+2}$ and $R \ge \log_4 n - d/2 - 1 \ge (\log_4 n)/2$. Multiply this tuple count by $\rho_d Q^d$ from (7) to obtain (2). □

For clarity, we give the lifting argument that transfers this result to a prescribed size in one more dimension. The argument is due to Wagner [4, Lemma 3.5].

Lemma 6. *For $d \ge 1$ and $n \ge 1$,*

$$C_{d+1}(n) \ge \sum_{j=1}^{n} C_d(j).$$

Proof. Take a j-point corner cut T in the plane $x_{d+1} = 0$, and choose a strict representation $w \cdot x < 1$. Within its open representation set we can choose the coordinates of w, together with one, to be linearly independent over $\mathbb{Q}$. For $u > 0$ consider

$$D_u = \{(x, z) \in \mathbb{Z}_{\geq 0}^{d+1} : w \cdot x + uz < 1\}.$$

Its intersection with $z = 0$ is always T. For $u > 1$, $D_u = T$. As u decreases to zero, points enter at the values $(1 - w \cdot x)/z$, with $x \in T$, $z \geq 1$. These entry values are distinct: equality for two pairs implies a rational linear relation among $1, w_1, \ldots, w_d$, and then forces both z and x to agree. There are only finitely many such values above any fixed positive bound, and the cardinality tends to infinity. Thus some open interval of u's gives cardinality exactly n, with no boundary point. Cuts with distinct sections T are different. □

Proof of Corollary 2. Apply Lemma 6 to $A_{d-1}(n)$, and then use Theorem 1. □

5 Consequences and limitations

Corollary 7. *For every fixed $d \geq 2$, there are $c_d, \eta_d > 0$ such that, for every sufficiently large N, at least $\eta_d N$ integers $j \leq N$ satisfy*

$$C_d(j) \geq c_d j^{d-1} (\log j)^{d-1}.$$

Proof. Choose $a, b > 0$ with $A_d(N) \geq aN^d (\log N)^{d-1}$ for large N and $C_d(j) \leq bj^{d-1} (\log j)^{d-1}$ for large j. The finitely many exceptional terms contribute $O_d(1)$. For $0 < c < a/4$, the sum over indices failing the claimed threshold is at most $cN^d (\log N)^{d-1} + O_d(1)$. Every other term is at most $bN^{d-1} (\log N)^{d-1}$. At least a fixed positive multiple of N terms are therefore required. □

There is an established geometric interpretation. For a cut D, put $\alpha_D = \sum_{x \in D} x$. For fixed size n, these vectors are distinct and each is a vertex of their convex hull. Indeed, weights strictly cutting off D make $w \cdot \alpha_D$ smaller than $w \cdot \alpha_E$ for every other n-point set E: each replacement point has greater weight than every removed point. Thus α_D is uniquely exposed. Conversely every vertex of the finite hull is one of its generators. The vertex count of the corner-cut polytope is consequently $C_d(n)$. This correspondence is known [1, 7]; our contribution is the quantitative lower count, not the dictionary. For generic experimental designs the same count describes their corner-cut models [7, Definition 7 and Theorem 2].

The cumulative result gives sharp-order growth on average and the improved exact-size bound in dimensions at least four. It supplies no asymptotic leading constant and no pointwise $C_3(n) \gg n^2 (\log n)^2$ statement. It also gives no new enumeration algorithm, no uniform claim as d grows with n, and no fixed-prime unranking complexity improvement.

6 Prior results and reproducibility

The following distinctions delimit the contribution being assessed. The unrestricted density of rectangular primitive integer matrices is known [5, 6]. The later paper repairs a defect in the earlier density proof. A centered-cube density alone does not assert the uniform estimate for our moving boxes; Lemma 3 gives a direct finite proof. We claim no new minor criterion or unrestricted density formula.

Chen, Feng, Liu and Wu [12, Theorem 1] give finite probability bounds for appending independent rows to a prescribed primitive matrix. Their random entries use a common interval of length at least the maximum prescribed entry. Our boxes have translated intervals

of different lengths, with their shortest length potentially much smaller than the largest entry. That theorem is not a direct substitute for Lemma 3. Prime-by-prime rank analysis and finite determinant cutoffs are established methods; our claimed contribution is their use in the multiscale corner-cut count.

Finite-grid threshold encodings and planar counts are related background [9, 10]. Their parameters are the ambient grid and, in the latter case, intersections of two halfplanes. The argument here additionally controls the number of included points, restricts the positive intercepts, and sums disjoint aspect ranges. A difference in definitions alone is not evidence of novelty.

The Hilbert zonotope refines state polyhedra and supports polynomial-time computation of universal Gröbner bases in fixed dimension [8, Proposition 3.2 and Theorem 3.5]. Its count is an upper bound for a common refinement, not the lower bounded-cardinality count proved here. Likewise the lattice-subspace covering bounds of Balko, Cibulka and Valtr [11, Theorems 3–5] concern another counting parameter. Neither result is being recast as a new theorem.

Elizalde and Galván [13, Section 8.2] discuss higher-dimensional pyramidal partitions and refer to the corner-cut growth literature. For this draft, the relevant results in their 2025 journal article are provisionally assumed unchanged from arXiv:2312.16353v1, as authorized for the project. The final-version comparison is recorded separately for later checking. This assumption does not establish priority.

Current priority status. The proof establishes the stated inequalities and their strict improvement over the inspected lower bound in (1). The literature audit has not established an unrestricted claim of priority or submission readiness. In particular, the complete 2002 journal text has not been inspected; the proofs were compared with the author's full thesis chapter. The 2007 grid-encoding paper was compared at the abstract level only. The detailed audit accompanies the source at `results/research/corner_cut_growth_audit.json`. This paragraph must be resolved before the draft is described as submission-ready.

Exact checks. The standard-library Python checker `scripts/run_corner_cut_growth_checks.py` uses independent determinants, modular Gaussian elimination and exact rational feasibility. It checks primitive-plane injectivity, the conditional residue bound, a nonprimitive-projection negative control, and a degenerate arrangement with 18 interior vertices and 48 chambers. It exhausts stated small boxes in dimensions two through four and uses deterministic samples in dimensions five and six. The explicit constants and 352 scale-rounding cases in dimensions two through twelve are checked with rational arithmetic. These computations can falsify steps; they do not prove the uniform theorem or its novelty.

Run, from the repository root,

```
python3 -B scripts/run_corner_cut_growth_checks.py
```

The report records fixtures, witnesses, the exact command and SHA-256 hashes of its sources in `results/research/corner_cut_growth_checks.json`. The proof above is independent of the computation. The other project kernels and public rank/unrank interface are unchanged.

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